

# Needleman-Wunsch Scoring Matrix

match=2 mismatch=-1 gap=-2

|   |     | A   | A   | C  | A  | C   | A   | C   | A   | A   | A   |
|---|-----|-----|-----|----|----|-----|-----|-----|-----|-----|-----|
|   | 0   | -2  | -4  | -6 | -8 | -10 | -12 | -14 | -16 | -18 | -20 |
| A | -2  | 2   | 0   | -2 | -4 | -6  | -8  | -10 | -12 | -14 | -16 |
| C | -4  | 0   | 1   | 2  | 0  | -2  | -4  | -6  | -8  | -10 | -12 |
| C | -6  | -2  | -1  | 3  | 1  | 2   | 0   | -2  | -4  | -6  | -8  |
| C | -8  | -4  | -3  | 1  | 2  | 3   | 1   | 2   | 0   | -2  | -4  |
| A | -10 | -6  | -2  | -1 | 3  | 1   | 5   | 3   | 4   | 2   | 0   |
| A | -12 | -8  | -4  | -3 | 1  | 2   | 3   | 4   | 5   | 6   | 4   |
| C | -14 | -10 | -6  | -2 | -1 | 3   | 1   | 5   | 3   | 4   | 5   |
| C | -16 | -12 | -8  | -4 | -3 | 1   | 2   | 3   | 4   | 2   | 3   |
| A | -18 | -14 | -10 | -6 | -2 | -1  | 3   | 1   | 5   | 6   | 4   |

# Sequence Statistics

| Metric         | Sequence A | Sequence B |
|----------------|------------|------------|
| Base pairs     | 10         | 9          |
| GC content (%) | 30.0       | 55.6       |